Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/01/2020
PI Name:	Govind, Shubha Ph.D.		
Project Title:	Does Spaceflight Alter the Virulence of a Natural Parasite of Drosophila?		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Developmental biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Invertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology (3) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sgovind@ccny.cuny.edu	**Fax:**	FY 212-650-8585
PI Organization Type:	UNIVERSITY	**Phone:**	212-650-8476
Organization Name:	City College of New York		
PI Address 1:	Biology Department		
PI Address 2:	160 Convent Ave		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10031-9101	**Congressional District:**	13
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	11/01/2014	**End Date:**	10/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/31/2021 per NSSC information (Ed., 8/4/2020) NOTE: End date changed to 10/31/2020 per F. Hernandez/ARC (Ed., 6/20/19) NOTE: End date changed to 10/31/2019 per F. Hernandez/ARC (Ed., 4/13/18) NOTE: End date changed to 10/31/2018 per F. Hernandez/ARC (Ed., 9/21/17) NOTE: End date changed to 10/31/2017 per NSSC information (Ed., 8/3/16)		
Key Personnel Changes/Previous PI:	September 2020 report: (1) Jennifer Chou (technical assistant) rejoined the project in July 2020. (2) PhD student Brian Wey is expected to defend his thesis on Oct 14, 2020.		
COI Name (Institution):	Bhattacharya, Sharmila Ph.D. (NASA Ames Research Center)		
Grant/Contract No.:	NNX15AB42G		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	The seminal discovery of Toll and Toll-NF-KB signaling as a central mediator of innate immune responses in the model insect Drosophila catalyzed similar discoveries in other organisms including innate immunity in humans. These findings led to an appreciation of the fundamental roles of innate immunity in human immune deficiency on the one hand, and immune health of insects on the other. In Drosophila larvae, primed, but quiescent hematopoietic progenitors are called into action from an organ with functional similarities to mammalian bone marrow. Blood cells signal the fat body (equivalent to the mammalian liver) to activate the expression of antimicrobial peptide genes (humoral immunity). Antimicrobial peptides efficiently combat deadly bacteria and fungi. The Leptopilina wasps are natural metazoan enemies of Drosophila and their attack kills fruit flies. Wasp attack (e.g., L. boulardi) activates both the humoral and cellular immune arms of larval hosts, each being dependent on Toll-NFKB signaling. Previous NASA-funded studies of Drosophila have shown that stresses in the space environment compromise hematopoietic development and modulate larval and adult innate immunity. We hypothesize that stimuli from the space environment will also alter the virulence of the parasitic wasps, and therefore their success on the fly hosts cultured in space. In Drosophila-wasp co-culture experiments using the NASA Fruit Fly Lab hardware over 30-days, we propose to (1) examine whether the fly or the wasp partner wins the arms race in space; (2) examine the effects of spaceflight on the anti-parasite response in fruit fly hosts; and (3) examine if virulence of “space-cultured” wasps is different from counterparts grown on Earth. While both L. boulardi and L. heterotoma parasites always succeed on the wild type, only the latter wasp species produces unique cell-killing activities residing in its virus-like particles. It also suppresses humoral immunity completely. We will examine if changes in the expression of VLP (virus-like particles) genes underlies changes in parasite virulence affected by spaceflight. Host defense and virulence will be inferred from a suite of other well-established assays of fly/host survival and measures of immune response.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Parasitic wasps of Drosophila are a powerful and increasingly well-characterized model for understanding longstanding questions about natural host/parasite interactions, allowing molecular genetic examination of both innate immune mechanisms in fly hosts and virulence mechanisms of their wasp parasites. Distant relatives of these parasitic wasps are used for biocontrol of agricultural pests worldwide. The "Earth benefits" impact of our research over the last year has been as follows: (a) Ground and space whole body transcriptomes of L. heterotoma and L. boulardi have been obtained. Whole body transcriptomic data have been submitted to NCBI (National Center for Biotechnology Information). We are in the process of identifying candidate genes for further analysis among the hundreds of genes that are differentially expressed in ground versus space conditions in for each species. (b) Identification and isolation of viable mutants of L. heterotoma with golden wing color (aurum). Gray wing is the normal color. A second visible mutant with an angular wing margin (mutant is named kona) has also been identified. The normal wing margin is round. To our knowledge, these two strains of L. heterotoma are the only living mutants of any species of the Leptopilina genus. (c) Over 10 students directly trained in the lab over last year. One technician trained and contributed to the project. Seven Federal Work Study students trained and contributed to the project. (d) Three papers were published in the last reporting period and two student presentations were made in conferences in fall 2019. These presentations helped to disseminate our research results.
Task Progress:	(a) Ground and space whole body transcriptomes of L. heterotoma and L. boulardi have been obtained. Whole body transcriptomic data have been submitted to NCBI (National Center for Biotechnology Information). We are in the process of identifying candidate genes for further analysis among the hundreds that are differentially expressed in ground versus space conditions in for each species. (b) Identification and isolation of viable mutants of L. heterotoma with golden wing color (aurum). Gray wing is the normal color. A second visible mutant with an angular wing margin (called kona) has also been identified. The normal wind margin is round. To our knowledge, these are the only living mutants of any species of this genus.
Bibliography Type:	Description: (Last Updated: 01/09/2024)
Abstracts for Journals and Proceedings	Arguelles J, Govind S, Singh S. "Structural analysis of LhetKNOT, a venom peptide of Drosophila Parasitoid, Leptopilina heterotoma, reveals potential role of heparin-binding structural motif in antimicrobial activity of Knottin proteins." 2019 Entomological Society of America Meeting, St. Louis, MO, November 17-20, 2019. 2019 Entomological Society of America Meeting, St. Louis, MO, November 17-20, 2019. Virtual Poster 149682. , Nov-2019
Abstracts for Journals and Proceedings	Wright B, Chou, Govind S. "Examining Virulence Changes in Mutant Strains of Parasitic Wasps of Drosophila." Annual Biomedical Research Conference for Minority Students (ABRCMS), Anaheim, California, November 13-16, 2019. Annual Biomedical Research Conference for Minority Students (ABRCMS), Anaheim, California, November 13-16, 2019. Abstract # C-232. , Nov-2019
Articles in Peer-reviewed Journals	Wey B, Heavner ME, Wittmeyer KT, Briese T, Hopper KR, Govind S. "Immune suppressive extracellular vesicle proteins of Leptopilina heterotoma are encoded in the wasp genome." G3 (Bethesda). 2020 Jan 7;10(1):1-12. https://doi.org/10.1534/g3.119.400349 ; PMID: 31676506; PMCID: PMC6945029 , Jan-2020
Articles in Peer-reviewed Journals	Gilbert R, Torres ML, Clemens R, Hateley S, Hosamani R, Wade W, Bhattacharya S. "Spaceflight and simulated microgravity conditions increase virulence of Serratia marcescens in the Drosophila melanogaster infection model." npj Microgravity. 2020 Feb 4;6(1):4. https://doi.org/10.1038/s41526-019-0091-2 ; PMID: 32047838; PMCID: PMC7000411 , Feb-2020

Articles in Peer-reviewed Journals	Tahimic CGT, Paul AM, Schreurs AS, Torres SM, Rubinstein L, Steczina S, Lowe M, Bhattacharya S, Alwood JS, Ronca AE, Globus RK. "Influence of social isolation during prolonged simulated weightlessness by hindlimb unloading." Front Physiol. 2019 Sep 13;10:1147. https://doi.org/10.3389/fphys.2019.01147 ; PMID: 31572207; PMCID: PMC6753329 , Sep-2019